LINMA2222 - Stochastic Optimal Control & RL

Optimal Portfolio Strategy

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UCLouvain

— Outline

1. Modeling and Analysis (Part I)

1. Modeling and Analysis (Part I)

Average-reward trading control:

$$\max_{\pi} \lim_{T\to\infty} \frac{1}{T} * \mathbb{E}\left[\sum_{t=0}^{T-1} r_t\right]$$

State: $x_t = (q_t, z_t^a, z_t^u)^\top$
Control: u_t
Noise: $\xi_t^a, \xi_t^p \sim \mathcal{N}(0,1)$
Constraint: $q_t \in [0,1]$

True dynamics

$$q_{t+1} = q_t + u_t$$

$$z_{t+1}^a = (1-\omega^a) z_t^a + \omega^a \sigma^a \xi_t^a$$

$$z_{t+1}^u = (1-\omega^u) z_t^u + \omega^u \beta^u u_t$$

Price / execution and P&L

$$p_{t+1} = p_t + z_t^a + z_t^u + \gamma^u u_t + \sigma^p \xi_t^p$$

$$\overline{p}_{t,t+1} = \theta p_t + (1-\theta) p_{t+1} + \theta \gamma^u u_t$$

Reward: Profit / Loss g_t

$$g_t := 1000 \cdot \left(\underbrace{(q_{t+1} p_{t+1} - q_t p_t)}_{\text{portfolio value change}} - \underbrace{(q_{t+1} - q_t)\overline{p}_{t,t+1}}_{\text{cash flow}} \right)$$

$$r_t := c(g_t) = \max\left(g_t - \frac{1}{2} g_t^2, 1 - \exp(-g_t)\right)$$

- g_t: Net profit/loss from trading
- **Change in portfolio value at market price**
- **Cash spent / received**

baseline policy

$$\pi_{\text{cl}(x_t)} = K_{\text{cl}} x_t = -0.5 q_t + 0.5 z_t^a + 0.5 z_t^u$$

$$\pi_{\text{pcl}(x_t)} = \max\Big(-q_t, \min\big(1 - q_t, \pi_{\text{cl}(x_t)}\big)\Big)$$

Average rewards:

- π_{cl} : **0.011451**
- π_{pcl} : **0.005796**

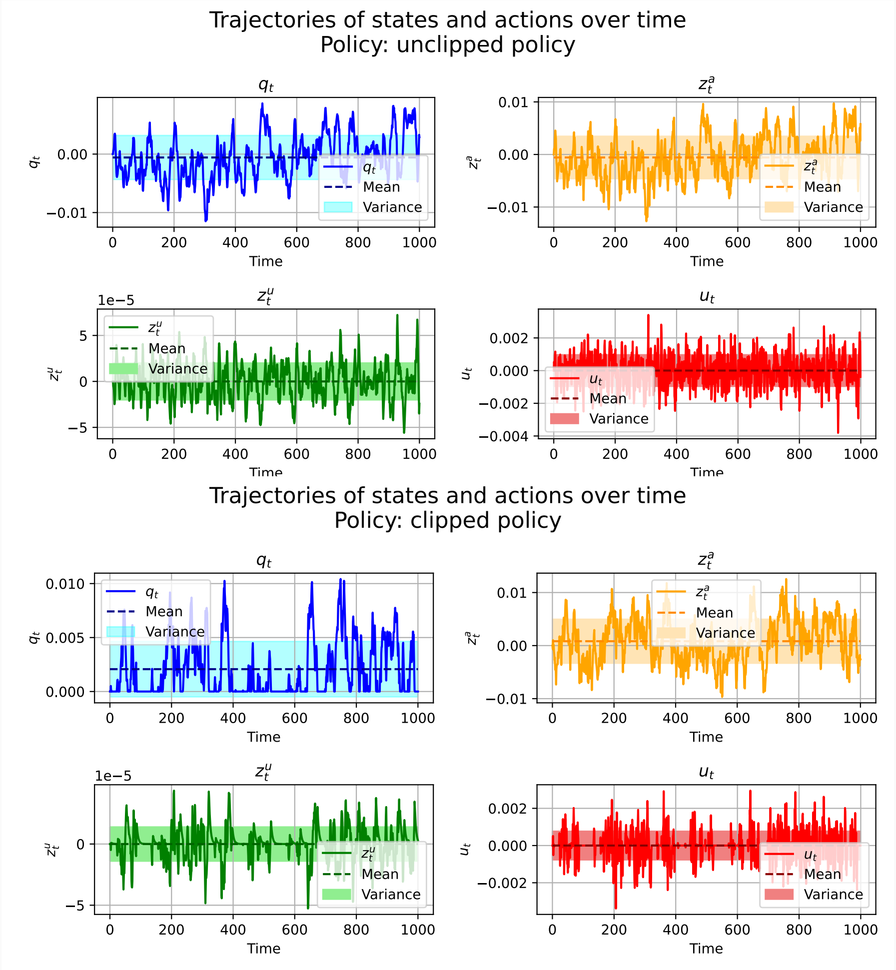

Figure 2: π_{cl} and π_{pcl} : Example trajectory

- can make $q_t < 0$

  => infeasible

CMA-ES linear policy

$$\pi_{l(x)} = \text{clip}(p_3^\top [q, z^a, z^u], [-q, 1-q])$$

CMA-ES quadratic policy

$$\pi_{q(x)} = \text{clip}(p_{10}^\top \varphi(x), [-q, 1-q])$$

where $\varphi(x) = \left[1, q, z^a, z^u, q^2, (z^a)^2, (z^u)^2, qz^a, qz^u, z^a z^u\right]^\top$

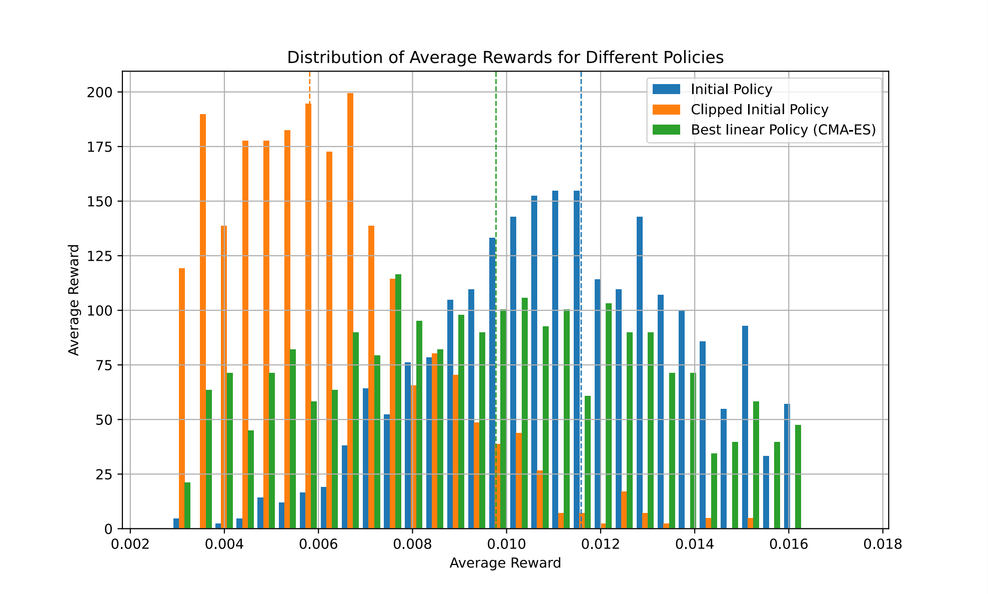

Figure 3: Reward comparison: baseline vs CMA-ES policie

Method made for noisy optimization.

- quadratic only slightly better.

2. LQR, LQG and MPC (Part II)

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LQR approximation

Linear dynamics $x_{t+1} = Fx_t + Gu_t + D\xi_t$

Reward: Maximize the expected reward (minimize cost):

$$\hat{r}(x,u) := \frac{1}{2}x^\top Q x + x^\top S u + \frac{1}{2}u^\top R u$$

$$\max \quad \lim_{T\to\infty} \frac{1}{T}\mathbb{E}\left[\sum_{t=0}^{T-1} \hat{r}(x_t, u_t)\right]$$

$$\mathbb{E}[r(x,u)] \approx \mathbb{E}\left[c_{\text{quad}}(g(x,u))\right] \approx \mathbb{E}[\hat{r}(x,u)]$$

with $c_{\text{quad}(g)} = g - \frac{1}{2}g^2$

Resulting matrices

$$F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-\omega^a & 0 \\ 0 & 0 & 1-\omega^u \end{pmatrix}$$

$$G = \begin{pmatrix} 1 \\ 0 \\ \omega^u \beta^u \end{pmatrix}$$

- linearize dynamics & quadratic reward
- obtain LQR model (F,G,Q,R,S)

LQR policy derivation

Discrete Algebraic Riccati Equation (DARE)

$$M^* = Q + F^\top M^* F - (F^\top M^* G + S)(R + G^\top M^* G)^{-1}(G^\top M^* F + S^\top)$$

Optimal gain

$$K_{\text{lqr}} = -(R + G^\top M^* G)^{-1}(G^\top M^* F + S^\top) \approx \begin{pmatrix} \mathbf{1.112} & \mathbf{-2.649} & \mathbf{-2.528} \end{pmatrix}$$

Theoretical average reward

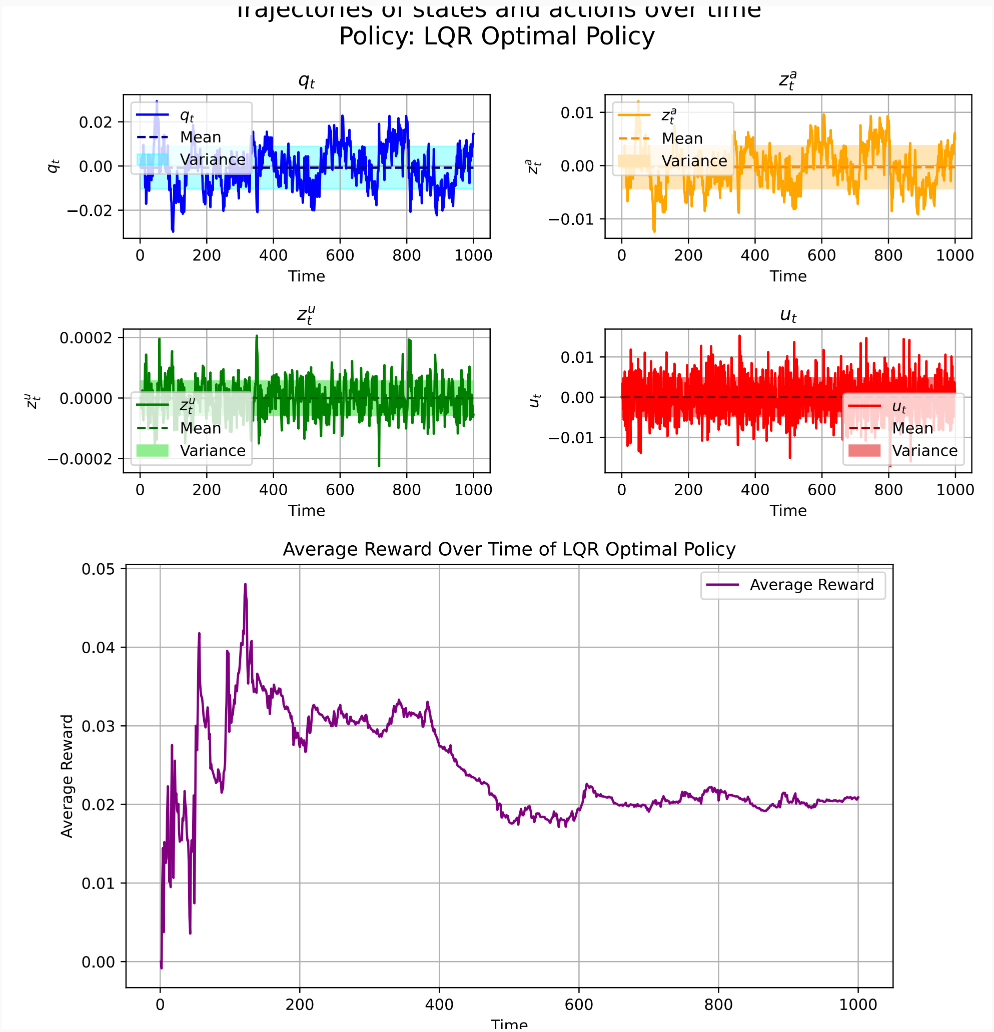
$$J^* = -\frac{1}{2}\operatorname{tr}(M^* D D^\top) = \mathbf{0.01936}$$

- Solve DARE iteratively or via scipy.linalg.solve_discrete_are
- Closed-loop stability: eigenvalues of $(F + GK)$ inside unit circle

LQR policy

$$\pi_{\mathrm{lqr}(x_t)} = -K_{\mathrm{lqr}} x_t$$

- average reward: **0.019467**
  - theoretical : **0.01936**
- average reward: **0.009693** (constrained)



Clipped LQR policy

- average reward decreases

Model Predictive Control

Variables: $z_k \in \mathbb{R}^{n_x}$ (predicted states), $v_k \in \mathbb{R}^{n_u}$ (predicted controls)

Optimization problem at time t:

$$\min_{z,v} \sum_{k=0}^{\mathcal{N}-1} \underbrace{\frac{1}{2} z_k^\top Q z_k + z_k^\top S v_k + \frac{1}{2} v_k^\top R v_k}_{\hat{r}(z_k, v_k)}$$

Subject to:

$$\begin{aligned} z_0 &= x_t \\ z_{k+1} &= F z_k + G v_k, & k &= 0, \dots, \mathcal{N}-1 \\ y_{\min} &\le z_{k+1} \le y_{\max}, & k &= 0, \dots, \mathcal{N} \\ u_{\min} &\le v_k \le u_{\max}, & k &= 0, \dots, \mathcal{N}-1 \end{aligned}$$

- z, q: decision variables (predicted trajectory)
- H, E: output matrix (here H selects position q_t)
- y_min, y_max: output bounds (position in [0,1])
- R_0: terminal cost (here R_0 = Q)

Stacked QP formulation

Stack variables:

$w = [z_0, \ldots, z_{\mathcal{N}}, v_0, \ldots, v_{\mathcal{N}-1}]^\top$

Solve :

$$\min_w \quad \frac{1}{2} w^\top H_{\text{obj}} w$$

$$H_{\text{obj}} = \begin{pmatrix} Q & & S & \\ & \ddots & & \ddots \\ & Q & & S \\ S^\top & & R & \\ & \ddots & & \ddots \\ & S^\top & & R \end{pmatrix}$$

Equality constraints :

$$A_{\text{eq}} w = [x_t, 0, \ldots, 0]^\top$$

bounds :

$$y_{\min} \le z_k \le y_{\max}, \quad u_{\min} \le v_k \le u_{\max}$$

- Standard QP: efficient solvers available
- Sparse structure exploited
- Receding horizon: only apply v_0^*

- express future states in terms of x_0 and controls V

State prediction: $z_k = F^k x_0 + \sum_{j=0}^{k-1} F^{k-1-j} G v_j$

In matrix form:

$$\underbrace{\begin{pmatrix} z_0 \\ z_1 \\ \vdots \\ z_N \end{pmatrix}}_{Z} = \underbrace{\begin{pmatrix} I \\ F \\ \vdots \\ F^N \end{pmatrix}}_{\mathcal{F}} x_0 + \underbrace{\begin{pmatrix} 0 & 0 & \dots & 0 \\ G & 0 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ F^{N-1}G & \dots & FG & G \end{pmatrix}}_{\mathcal{G}} \underbrace{\begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{pmatrix}}_{V}$$

Reduced QP: optimize $V \in \mathbb{R}^{n_u \cdot N}$

$$\min_V \quad \frac{1}{2} V^\top \hat{H} V + \left(\hat{F} x_0\right)^\top V$$

$$\hat{H} = \mathcal{G}^\top \tilde{Q} \mathcal{G} + \tilde{R} + \mathcal{G}^\top \tilde{S} + \tilde{S}^\top \mathcal{G}$$

$$\hat{F} = \mathcal{G}^\top \tilde{Q} \mathcal{F} + \tilde{S}^\top \mathcal{F}$$

where $\tilde{Q}$, $\tilde{R}$, $\tilde{S}$ are block-diagonal

Constraints:
input bounds + state bounds via $\mathcal{G}$

2. LQR, LQG and MPC (Part II)
— MPC: Condensed Formulation

- States eliminated $\longrightarrow$ QP size reduced from $(n_x + n_u) \cdot N$ to $n_u \cdot N$
- $\hat{H}$ precomputed once; only $\hat{F} x_0$ updated each step

chosen horizon : $\mathcal{N} = 10$

- performance vs computation
- greater N : diverging predictions

Policy	Avg Reward	Feasible	Key characteristic
π_{lqr}	0.019467	✗	optimal for LQR model
Clipped π_{lqr}	0.009693	✓	still good performance
MPC ($\mathcal{N} = 30$)	0.0113082	✓	2x slower than $\mathcal{N} = 10$
MPC ($\mathcal{N} = 10$)	0.0113081	✓	costly
MPC ($\mathcal{N} = 5$)	0.0113076	✓	$\approx$ time $\mathcal{N} = 10$

Takeaways:

- Constraints significantly reduce reward
- Clipped $\pi_{\text{lqr}} \approx$ MPC performance, but **much faster**

- In practice:
  - clipped LQR
    - fast and good performance

3. Policy Improvement (Part III)

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E-PIA on LQR model

Policy iteration steps

1. **Evaluation:** Solve Lyapunov equation for P_k:

$$P_k = Q_k + A_k^\top P_k A_k$$

where $A_k = F + GK_k$, $Q_k = Q + SK_k + (SK_k)^\top + K_k^\top R K_k$

2. **Improvement:** Update gain:

$$K_{k+1} = -(R + G^\top P_k G)^{-1}(S^\top + G^\top P_k F)$$

3. Policy Improvement (Part III)
— 3.1 Exact Policy Iteration Algorithm (E-PIA) (Q5)

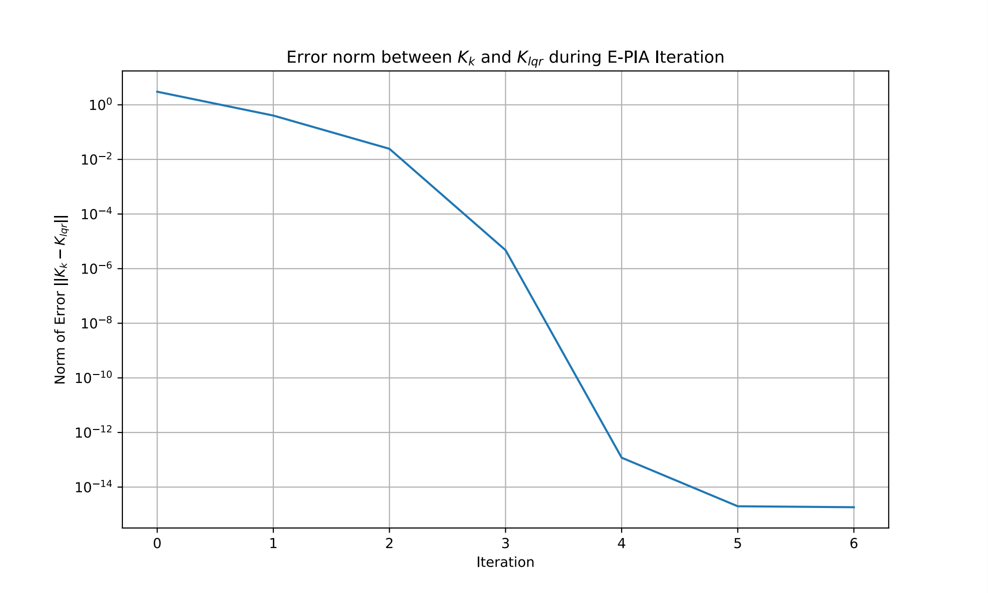
E-PIA convergence

Results

- From π_{cl}, converges to K_{lqr} in ≈ 5 iterations
- Confirms that Riccati-based LQR is optimal for the quadratic model

Learned gain

$$K_{\mathrm{lqr}} \approx \begin{pmatrix} 1.112 & -2.649 & -2.528 \end{pmatrix}$$



3. Policy Improvement (Part III)
— E-PIA Results (Q5)

LSTD for policy evaluation

Objective Approximate the Q-function under a fixed policy π:

$$Q^\pi(x,u) \approx \theta^\top \psi(x,u)$$

LSTD update Solve for θ using instrumental variables:

$$\theta = \left[\frac{1}{N} W + R_N\right]^{-1} \phi_N$$

where:

$$R_N = \frac{1}{N}\sum_{k=1}^{N} \Upsilon_k \Upsilon_k^\top \qquad \phi_N = \frac{1}{N}\sum_{k=1}^{N} \Upsilon_{k+1}\gamma_k$$

1. $\Upsilon_k = \psi(x(k), u(k)) - \psi(x(k+1), \pi(x(k+1)))$
2. $\gamma_k = c(x(k), u(k))$

LSTD for true model

Key design choices

- **Degree-4 polynomial basis**: cost contains exponential term
- **No bias term**
- **Regularization**: $(R_N + \varepsilon I)$ for numerical stability
- **Large N**: $N = 3000$ samples for well-conditioned matrix

Results

- Good match with Monte-Carlo estimates on true system

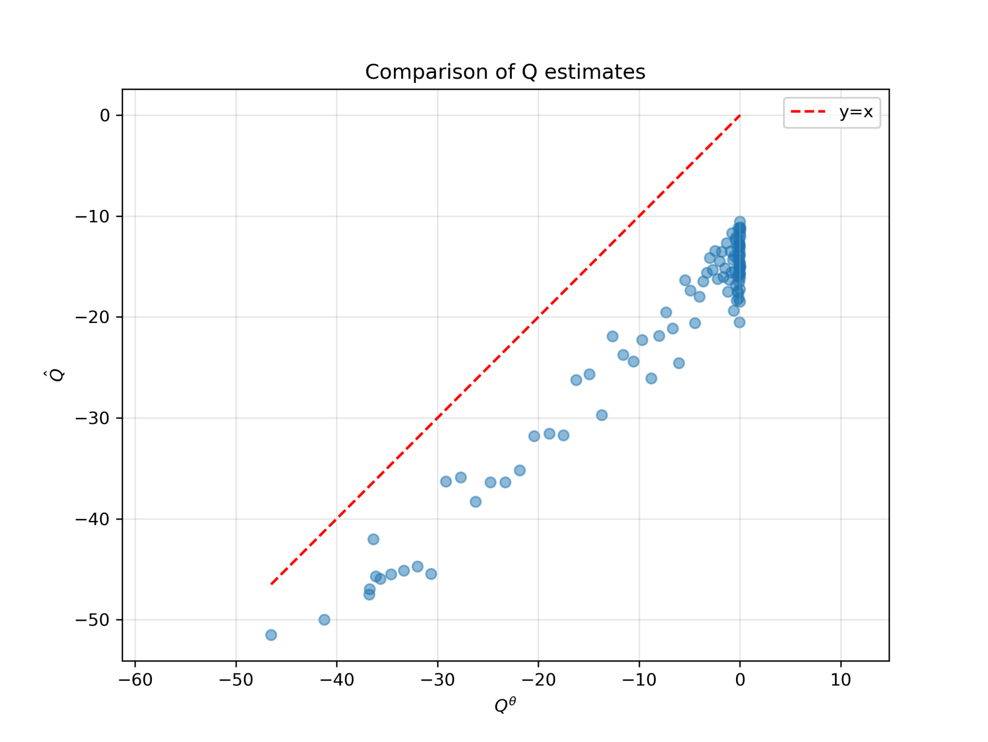


Figure 7: True system: Q^θ vs $\hat{Q}_{\mathrm{MC}}$

3. Policy Improvement (Part III)
— LSTD: Design Choices & Results

LSTD for LQR model

Key design choices

- **Degree-2 polynomial basis**: Q_{exact} is quadratic

Results

- Perfect match with the exact Q-function

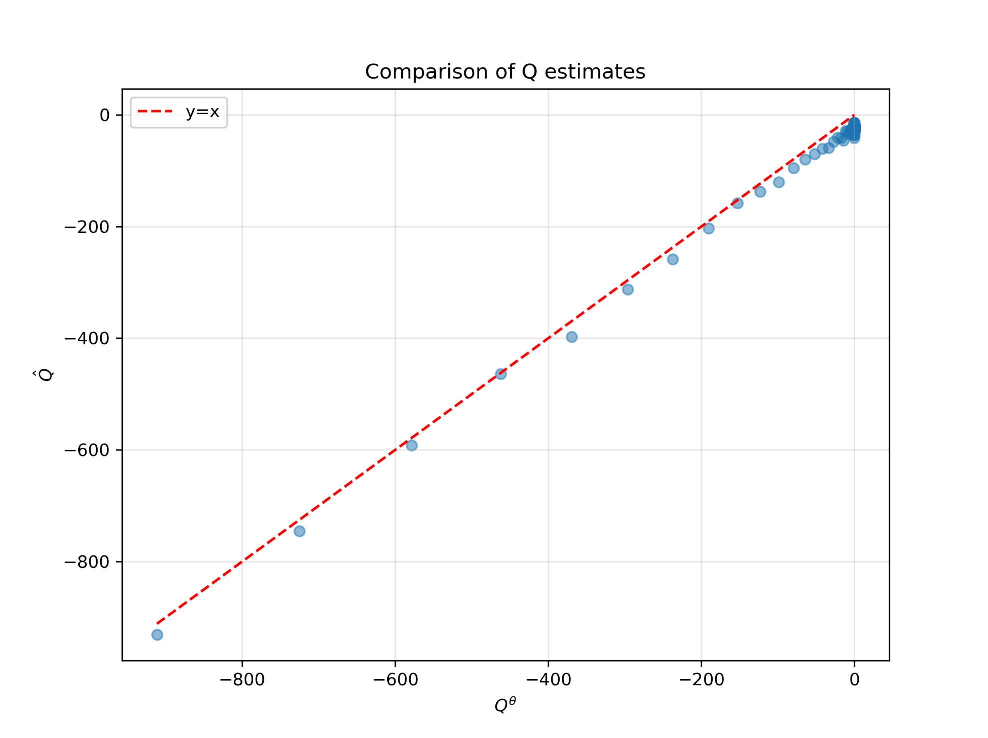


Figure 8: LQR model: Q^θ vs Q_{exact}

3. Policy Improvement (Part III)
— LSTD: Design Choices & Results

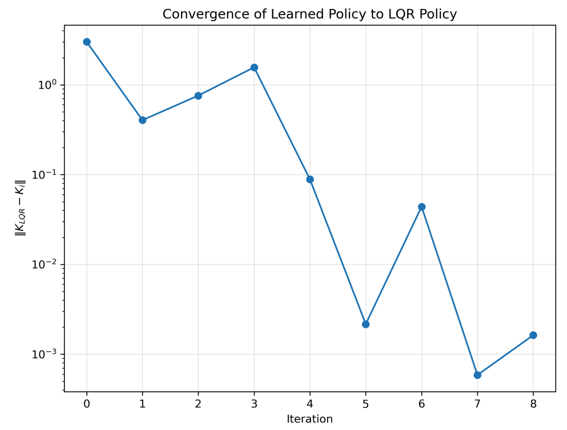
LSPI algorithm

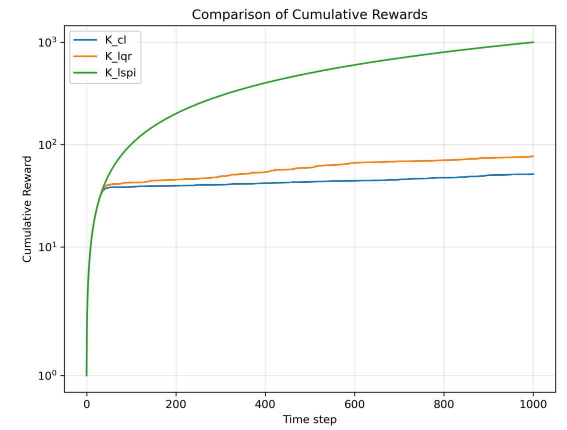
Iterate:

1. **Evaluation:** LSTD $\longrightarrow \theta_k$
2. **Improvement:** Find greedy policy w.r.t. Q^{θ_k}

Policy improvement step For linear policy $u = -Kx$:

- Sample states $x_s \in [-1, 1]^3$
- Optimize: $u_s^* = \arg\max_u \theta^\top \psi(x_s, u)$
- Fit: $K_{k+1} = (X^\top X)^{-1} X^\top (-U^*)$





Q-λ: online temporal-difference

Algorithm

- Quadratic features: $\psi(x, u) = \text{vec}([x; u][x; u]^\top)$
- $\zeta_{t+1} = \lambda \zeta_t + \psi_t$
- TD error: $\mathcal{D}_t = c_t + Q_{t+1} - Q_t$
- Update: $\theta \leftarrow \theta + \alpha \mathcal{D}_t \zeta_t$

Effect of λ

- $\lambda = 0$: one-step TD (slow but stable)
- $\lambda \to 1$: Monte-Carlo-like (faster but can diverge)

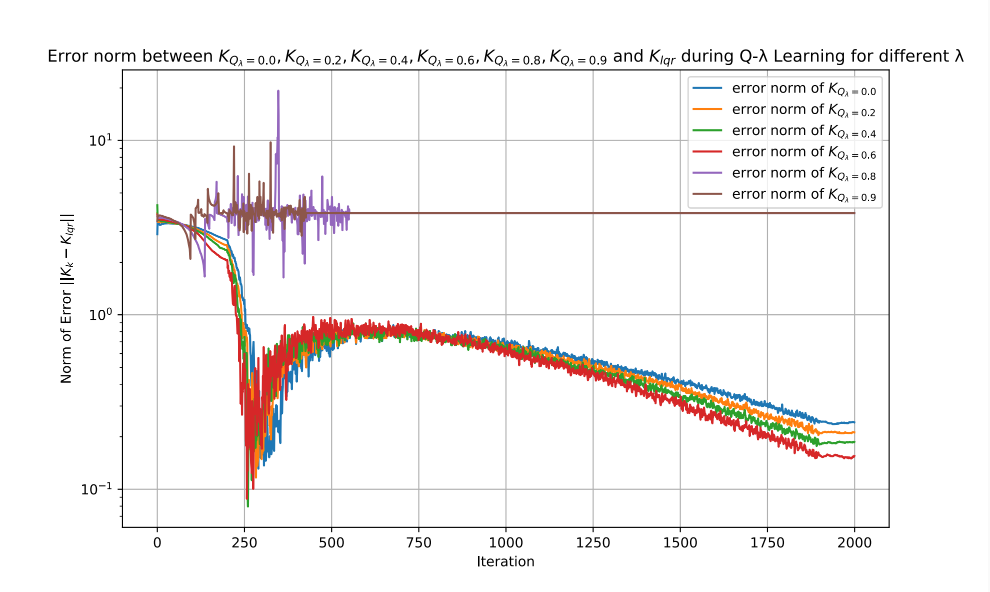


Figure 10: Convergence to K_{lqr} for different λ

3. Policy Improvement (Part III)
— 3.4 Q-λ Learning (Q7)

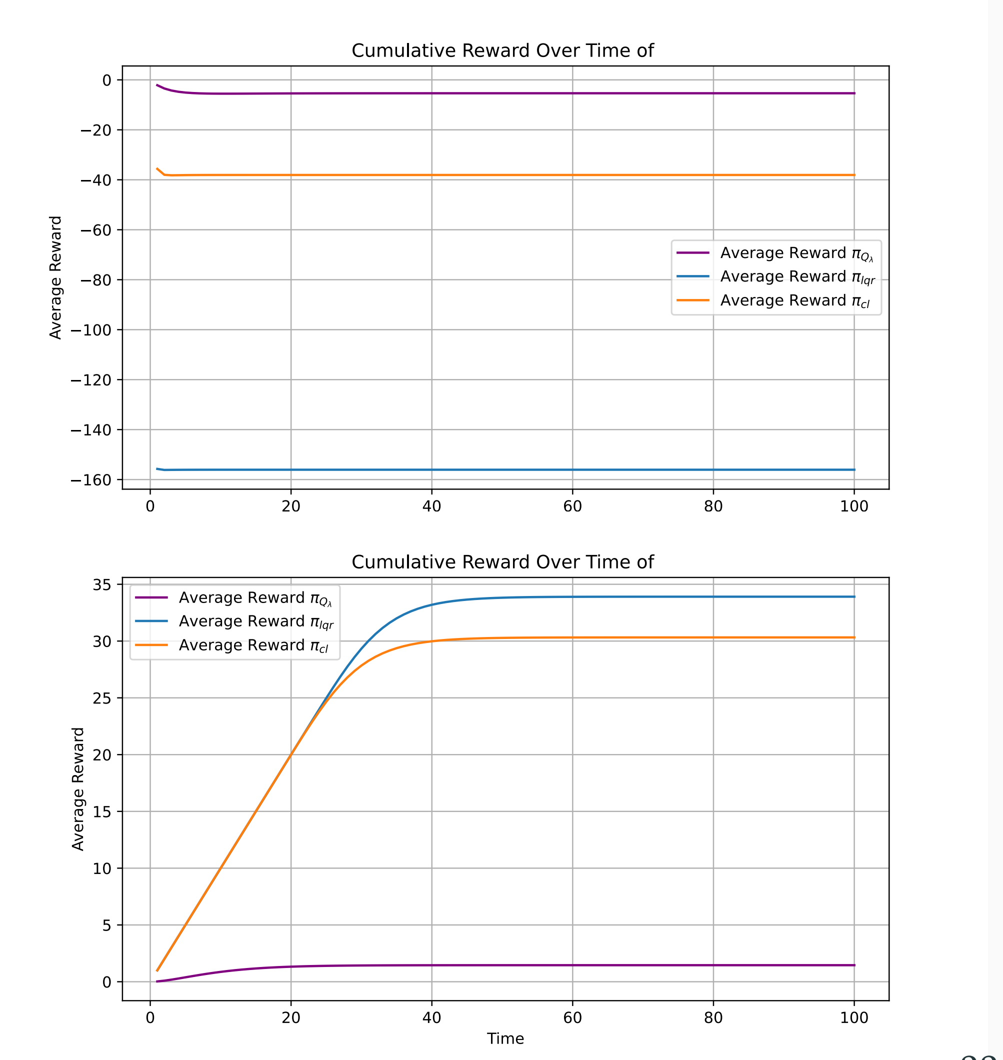
Q-λ performance

Convergence

- Converges to K_{lqr} for LQR settings
- Higher $\lambda \longrightarrow$ faster convergence (with tuned α)

Reward for deterministic system

- $\pi_{Q(\lambda)}$ less risky
  $\Longrightarrow$ less profit in some cases



3. Policy Improvement (Part III)
— Q-λ Results

Method	Model needed	Online	Key insight
E-PIA	Yes (full)	No	Exact convergence to K_{lqr} via Lyapunov + improvement
LSTD	No	No	Basis: degree-4 for true, degree-2 for LQR
LSPI	No	No	Iterate: LSTD evaluation + greedy improvement
Q-λ	No	Yes	λ controls bias-variance: higher $\lambda \longrightarrow$ faster but needs smaller α

Takeaways

- E-PIA: model-based benchmark, converges in 6 iterations
- LSTD/LSPI: model-free, basis choice critical (degree-4 for true reward)
- Q-λ: online learning, π_{Q_λ} is more conservative (less risky, less profit)

3. Policy Improvement (Part III)
— Part III Summary

4. LSPE for Average-Reward (Part IV)

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Why LSPE for average-reward?

Discounted setting ⟶ Bellman error

$$\mathcal{B}^\theta(x,u) = \mathbb{E}\left[r + \gamma Q^\theta(x', \pi(x')) \mid x, u\right] - Q^\theta(x,u)$$

Average-reward setting ⟶ Poisson error

$$\mathcal{P}^\theta(x,u) = \mathbb{E}\left[r + Q^\theta(x', \pi(x')) \mid x, u\right] - \eta^\theta - Q^\theta(x,u)$$

Key difference

- No discount factor $\gamma \longrightarrow$ need to subtract average reward η
- LSPE minimizes mean squared Poisson error
- Jointly estimates θ (Q-parameters) and η (average reward)

4. LSPE for Average-Reward (Part IV)
— 4.1 Motivation: Poisson vs Bellman Error

Least-Squares Policy Evaluation

Objective Find $\theta = [\theta_Q; \eta]$ that minimizes:

$$\min_{\theta} \mathbb{E}\left[\left(\mathcal{P}^{\theta}(x,u)\right)^2\right]$$

LSPE solution Define $\varphi_k = [\mathbb{E}[\psi'] - \psi_k; -1]$, then solve:

$$A\theta = b \quad \text{where} \quad A = \mathbb{E}[\varphi\varphi^{\top}], \quad b = -\mathbb{E}[\varphi \cdot r]$$

Implementation

- Monte-Carlo approximation of $\mathbb{E}[\psi(x', \pi(x'))]$
- Regularization: $(A + \lambda I)$ for stability
- Quadratic basis ψ: 14 features (no constant term)

4. LSPE for Average-Reward (Part IV)
— 4.2 LSPE Algorithm (Q8.5–Q8.6)

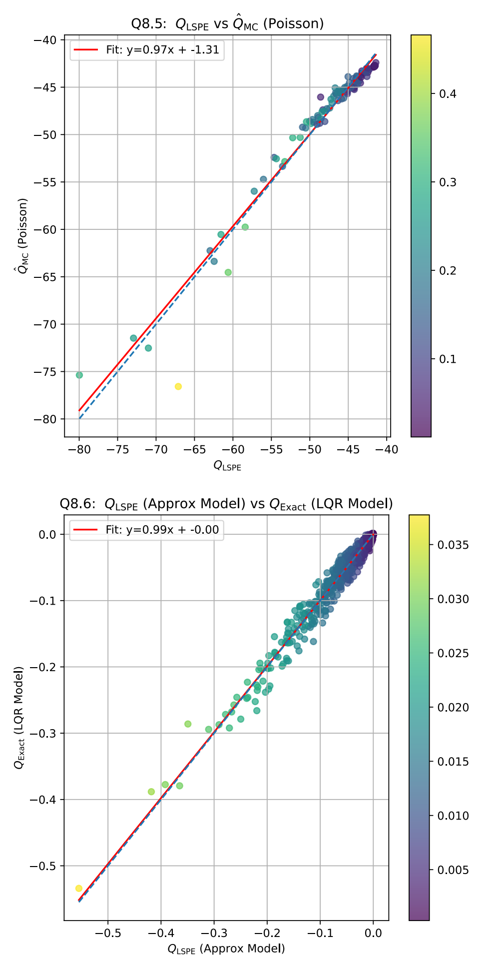
Q8.5–Q8.6: Evaluating π_{cl}

Q8.5: True system

- LSPE matches Monte-Carlo $\hat{Q}$
- Validates the Poisson formulation

Q8.6: LQR approximate model

- LSPE matches exact Q_{LQR}
- Confirms correctness of implementation



LSPE+PI algorithm

Iterate:

1. **Data collection:** Explore with current policy + noise ($\sigma_{\text{exp}} = 0.02$)
2. **Evaluation:** LSPE $\longrightarrow \theta_Q, \eta$
3. **Improvement:** Greedy policy from Q^θ

Greedy policy extraction For quadratic $Q(x, u) = \ldots + b(x)u + au^2$:

$$u^* = -\frac{b(x)}{2a}$$

Two settings tested

- Q8.7: Train & evaluate on **true system**
- Q8.8: Train on **LQR model**, evaluate on true system

4. LSPE for Average-Reward (Part IV)
— 4.3 LSPE + Policy Iteration (Q8.7–Q8.8)

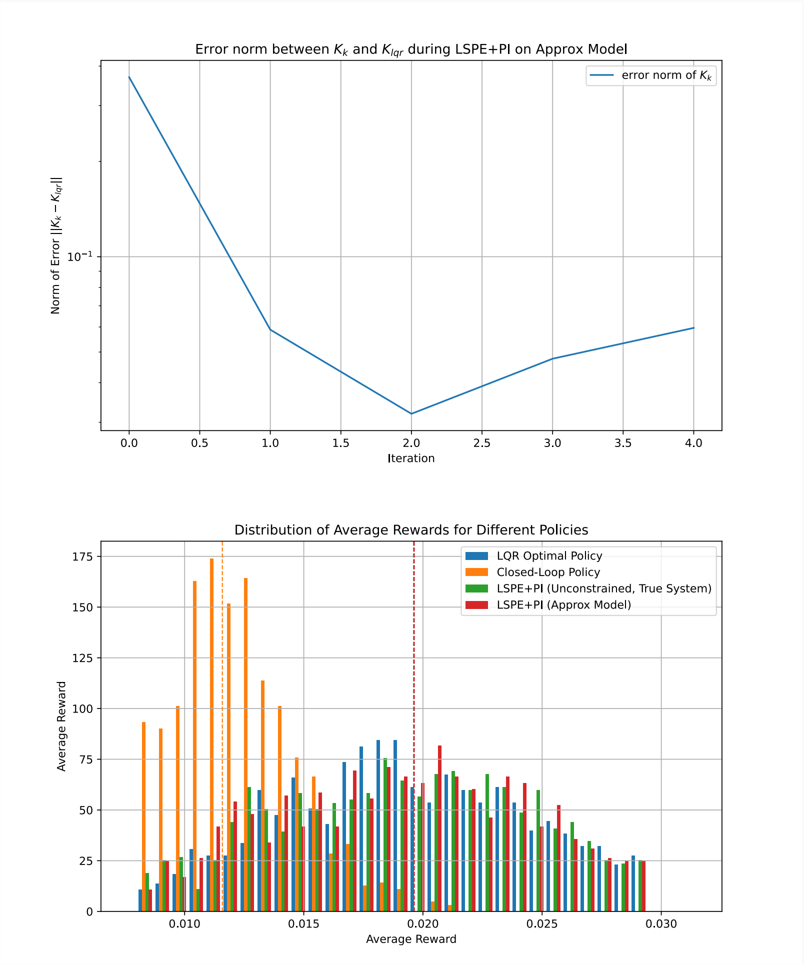
Convergence and performance

Q8.8: LQR model

- Gain K converges to K_{lqr} in *few* iterations
- Confirms LSPE+PI recovers optimal LQR policy

Q8.7: True system

- Similar performance as K_{lqr}



4. LSPE for Average-Reward (Part IV)
— LSPE+PI Results (Q8.7–Q8.8)

Enforcing $q \in [0, 1]$

Constrained improvement

$$u^* = \arg \max_{u \in [-q, 1-q]} Q^\theta(x, u)$$

Effect of constraints

- Saturation episodes reduce reward
- Unconstrained policies look better... but are infeasible!

Fair comparison

- Compare all policies under same constraint

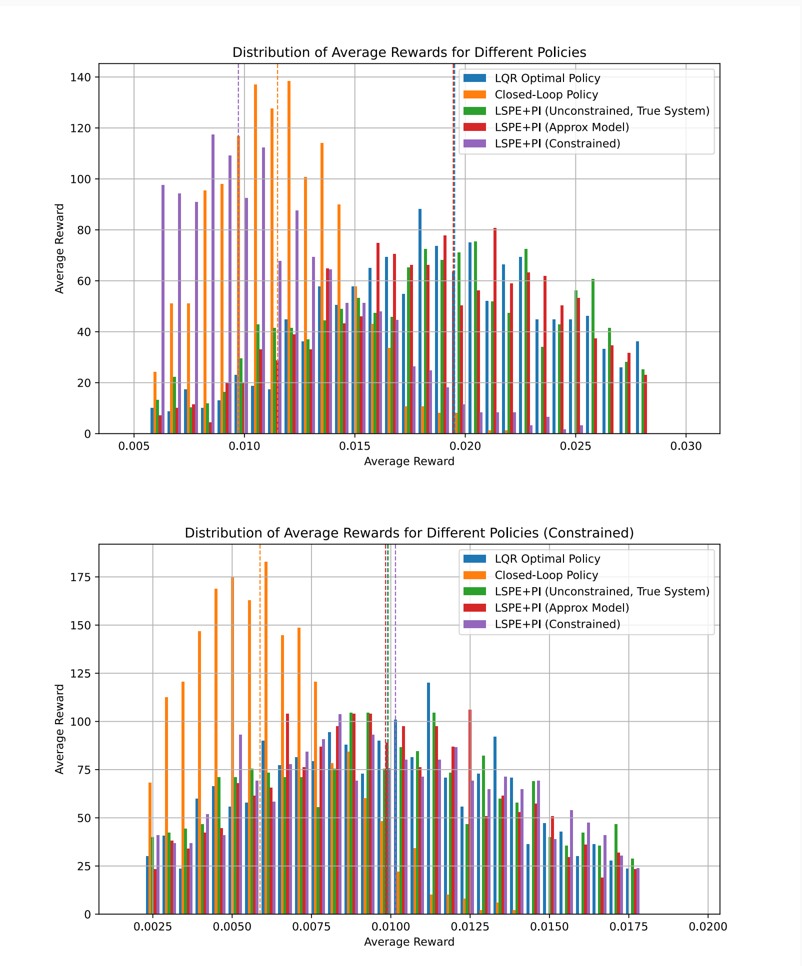


Figure 14: Unconstrained (above) & Constrained (below)

Question	Setting	Key result
Q8.5	True system	$Q_{\text{LSPE}} \approx \hat{Q}_{\text{MC}}$ — validates Poisson formulation
Q8.6	LQR model	$Q_{\text{LSPE}} \approx Q_{\text{exact}}$ — confirms implementation
Q8.7	LSPE+PI, true	Converges to near-optimal policy
Q8.8	LSPE+PI, LQR	Recovers K_{lqr} exactly
Q8.9	Constrained PI	Saturation reduces reward; fair comparison needed

Takeaways

- LSPE naturally handles average-reward setting via Poisson error
- Joint estimation of θ_Q and η (average reward)
- Constraint handling requires care in both improvement and evaluation

5. Model Comparison

5. Model Comparison

Policy	Model	const	Avg Reward	Avg Reward (unc)	Comments
π_{cl}	True	✗	0.005796	0.011451	
CMA-ES	True	✓	0.009654	—	
π_{lqr}	**LQR**	✗	0.009693	0.019467	$\to K_{\text{lqr}}$
MPC ($\mathcal{N} = 10$)	**LQR** det	✓	0.011308	—	higher compute
E-PIA	**LQR**	✗	0.009651	0.019584	$\to K_{\text{lqr}}$
π_{lspi}	True	✗	—	—	near-LQR on true model
Q_λ	True det	✓	0.000011	0.000019	
π_{lspepi} approx	**LQR**	✗	0.009833	0.019534	$\to K_{\text{lqr}}$
π_{lspepi} true	True	✗	0.009763	0.019452	$\to K_{\text{lqr}}$
π_{lspepi} true constr	True	✓	0.009725	—	

COMPUTED on True system :

- $T = 200$
- $N = 5000$ ($N = 100$ for MPC)
- $x_0 = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$

Comparison :

- mpc is better (but tested only at N=100)
- π_{lqr}, Q_λ on LQR model all methods converge to Klqr
- method trained with quadratic psi tend to ≈ LQR performance :
  - models rewards small variance => near zero (near our LQR approximation with c_quad)

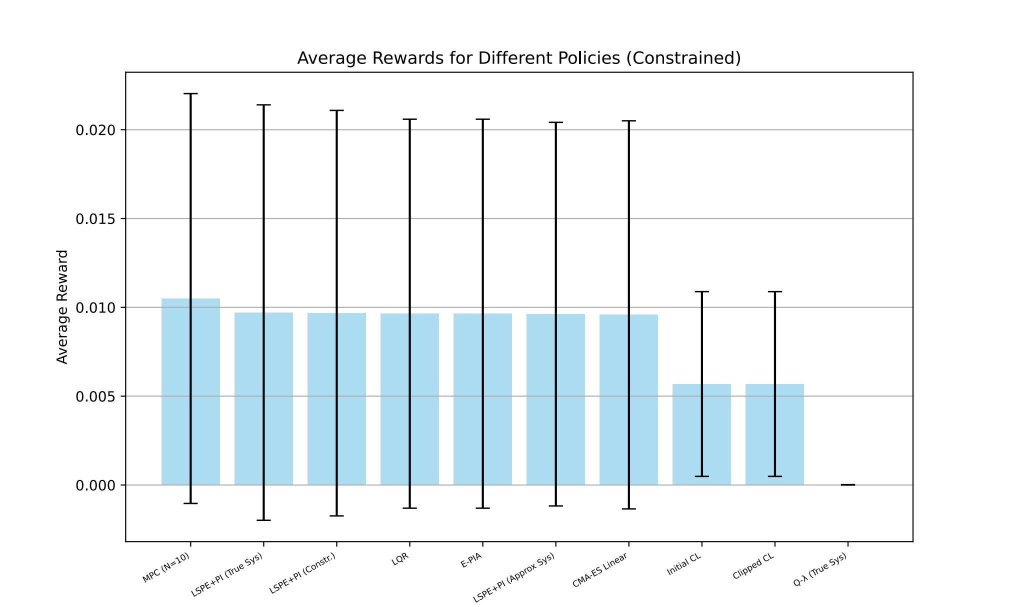


Figure 15: Average reward comparison

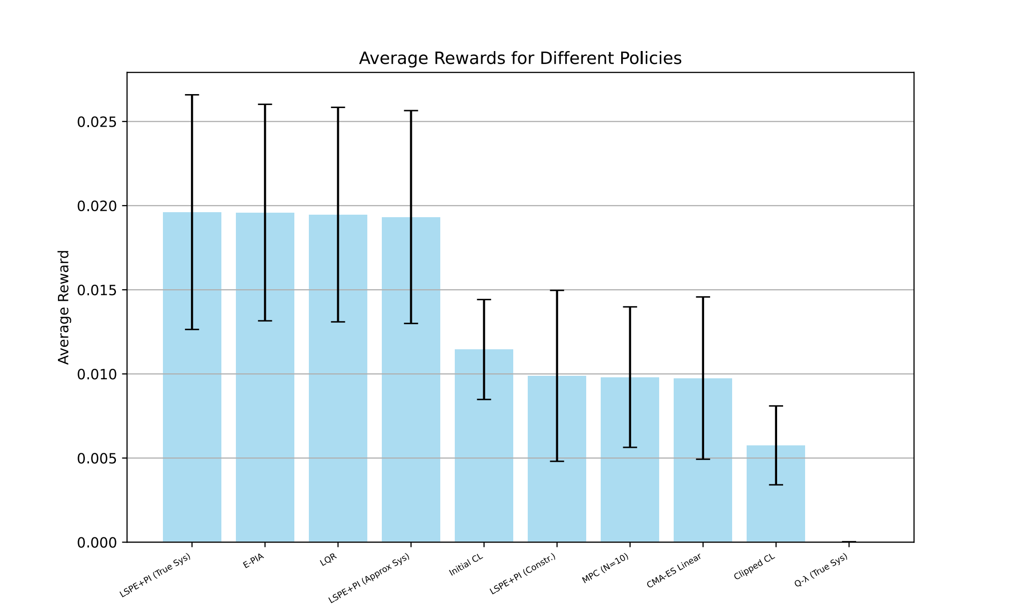


Figure 16: Average reward comparison (unconstrained)

MPC reward computed on $N = 200$ samples due to high compute cost

Design choices & difficulties

Design choices

- enforce feasibility via clipping; compare with MPC
- CMA-ES for noisy/non-smooth policy optimization
- quadratic features for LQR-consistent learning; higher-degree for true utility
- MPC horizon $\mathcal{N} = 10$ as compute/reward trade-off
- exploration: Gaussian around $K_{\text{cl}}x$ with tuned variance

Difficulties + how we addressed them

- saturation episodes reduce reward $\Longrightarrow$ compare "fairly" (all constrained vs all unconstrained)
- Q-λ instability at large $\lambda \Longrightarrow$ reduce learning rate α
- average-reward evaluation needs $\eta \Longrightarrow$ Poisson/LSPE directly estimates it

Thank you for your attention !

Aymeric Couplet and Lucas Ahou

5. Model Comparison
— The end

6. Appendix

6. Appendix

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Baseline distributions

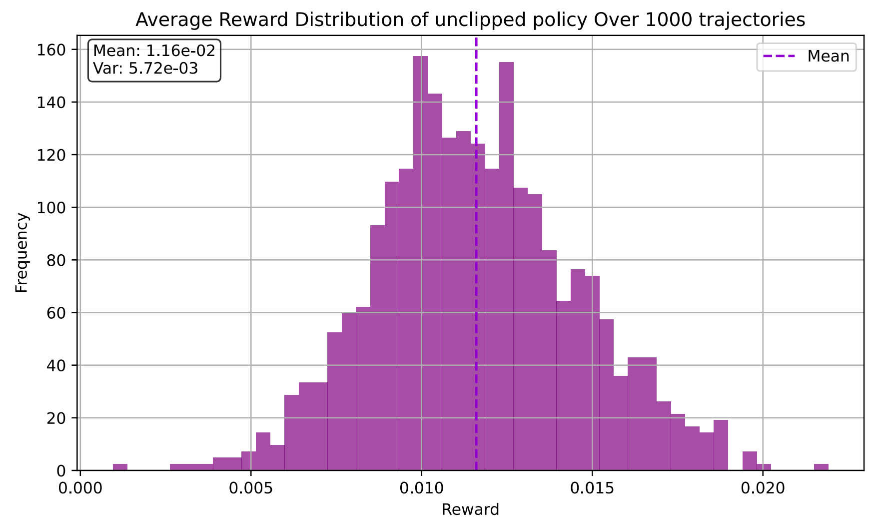

Figure 17: Unclipped baseline

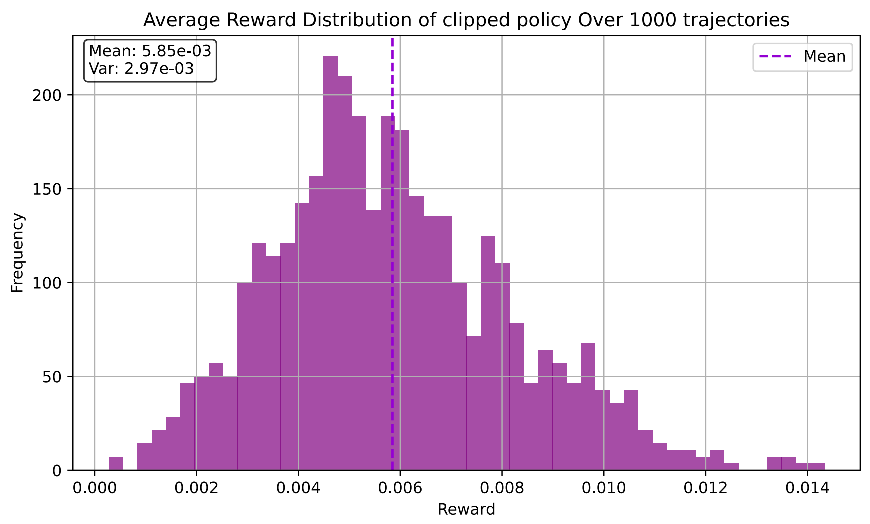

Figure 18: Clipped baseline

LQR distributions

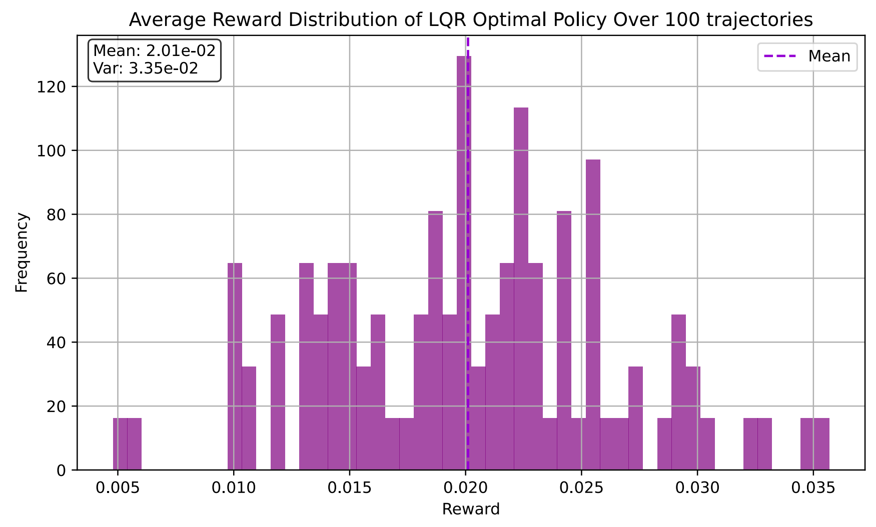

Figure 19: Unclipped LQR

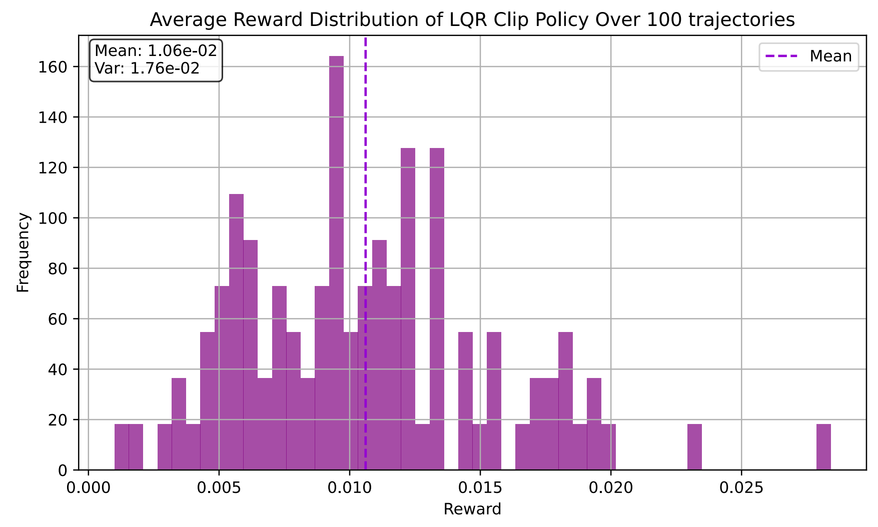

Figure 20: Clipped LQR

MPC extra plots

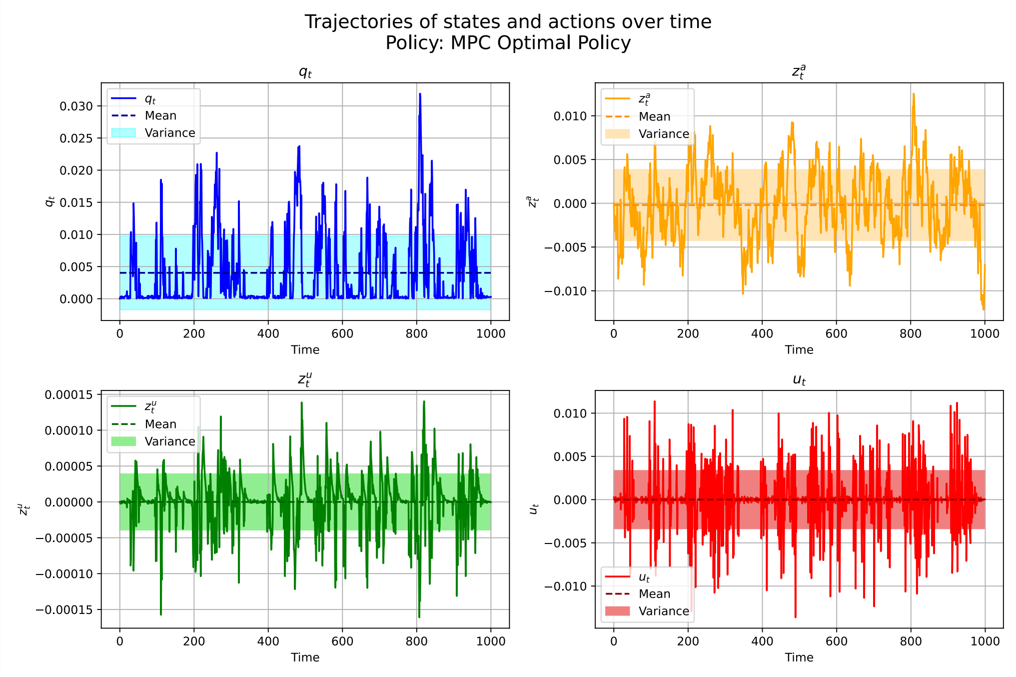

Figure 21: Example MPC states/actions

6. Appendix
— Appendix: MPC trajectories (Q4.10–Q4.11)

E-PIA convergence

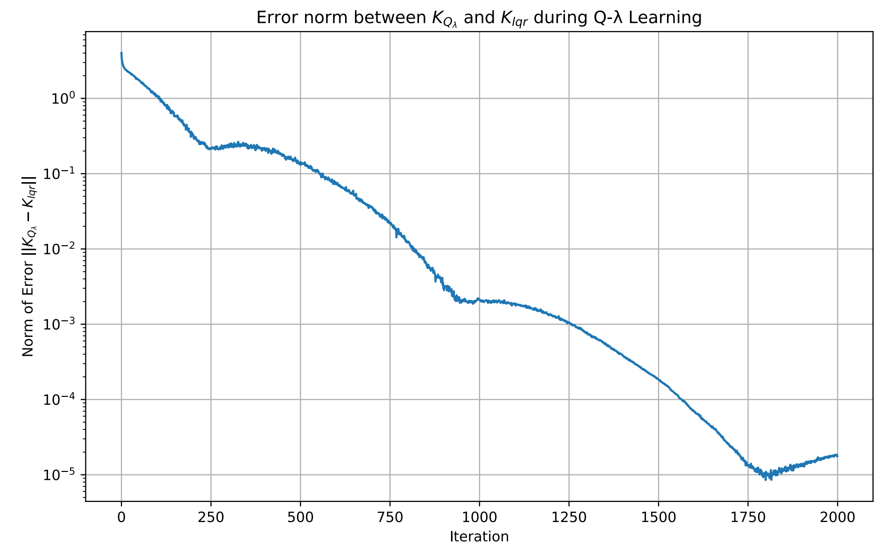

Figure 22: Error norm to K_{lqr}

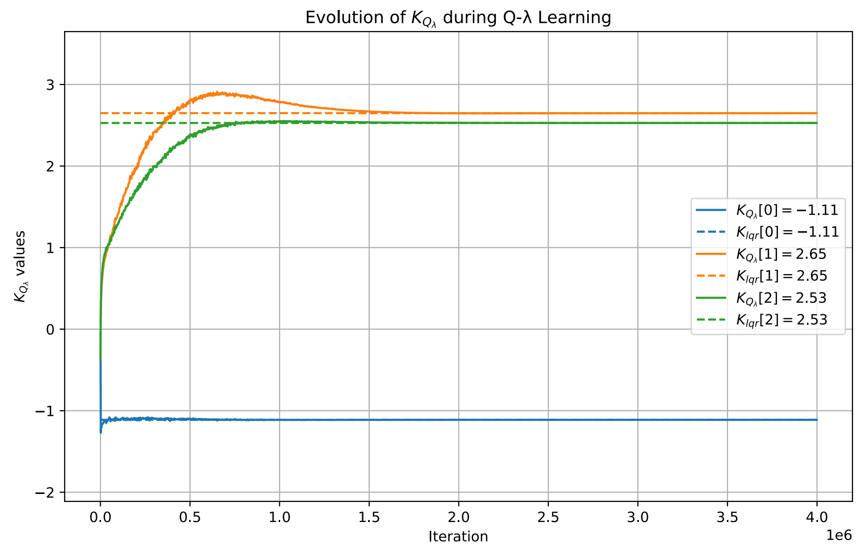

Figure 23: K_k components

UCLouvain

LSTD evaluation

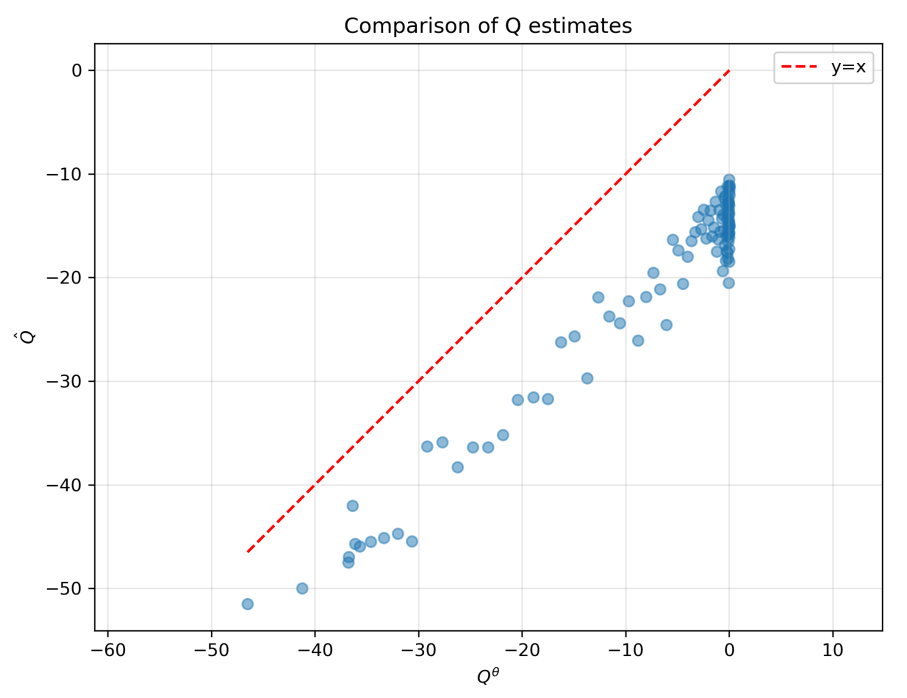

Figure 24: True system: Q^θ vs $\hat{Q}$

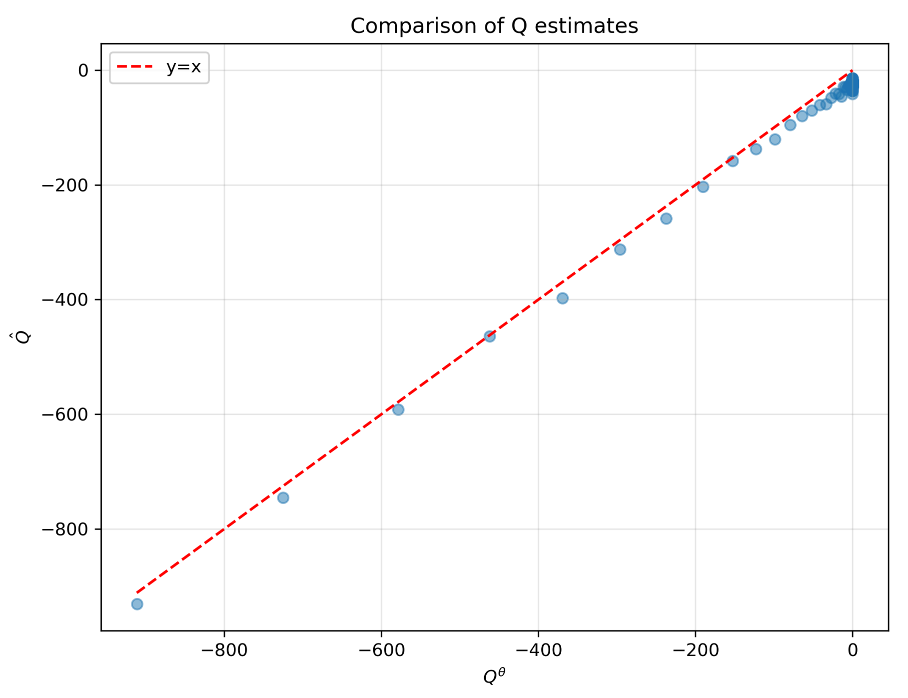

Figure 25: LQR model: Q^θ vs $\hat{Q}$

6. Appendix
— Appendix: LSTD evaluation plots (Q6.3–Q6.4)

LSPI: convergence + constraints

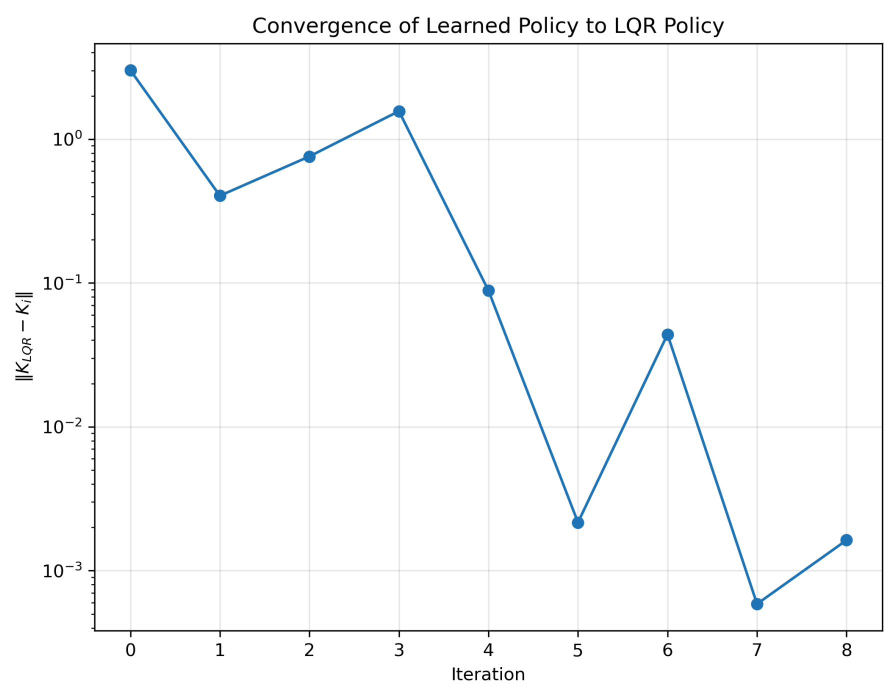

Figure 26: LSPI gain converges to K_{lqr}

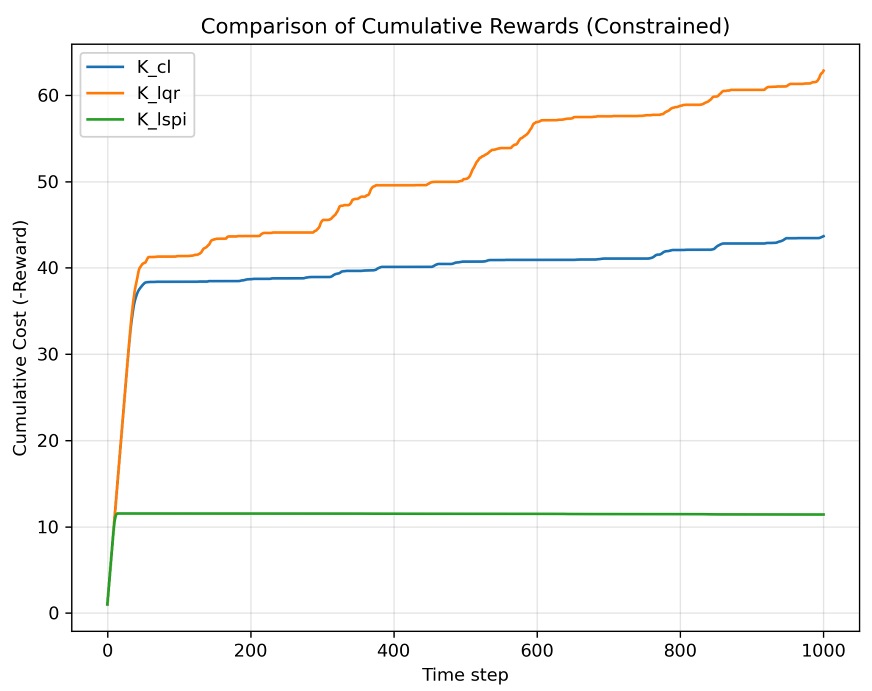

Figure 27: Constrained PI can reduce reward

LSPE+PI with constrained improvement

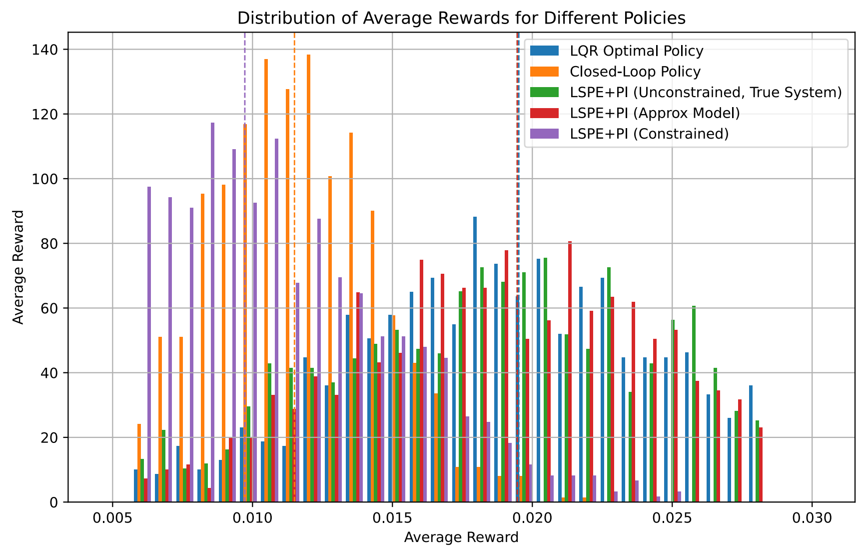

Figure 28: Unconstrained comparison

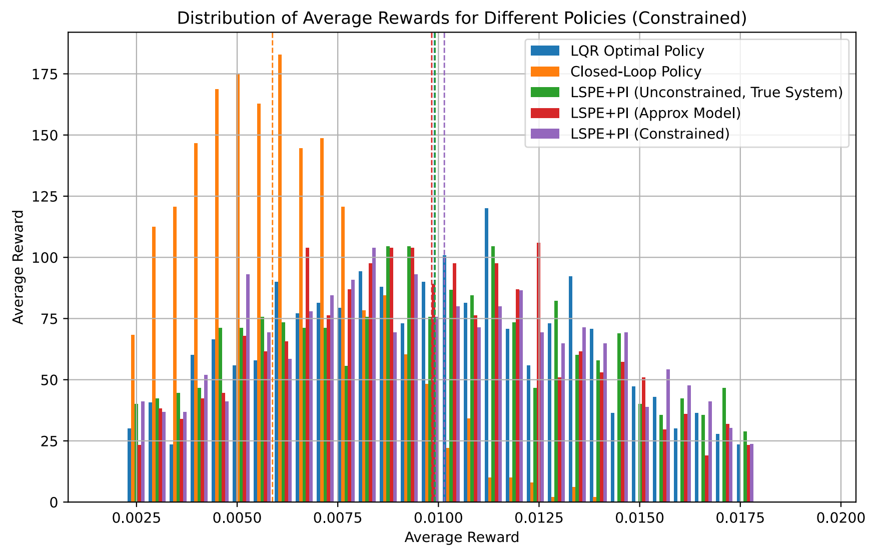

Figure 29: Fair comparison: all constrained

6. Appendix
— Appendix: LSPE constrained comparison (Q8.9)

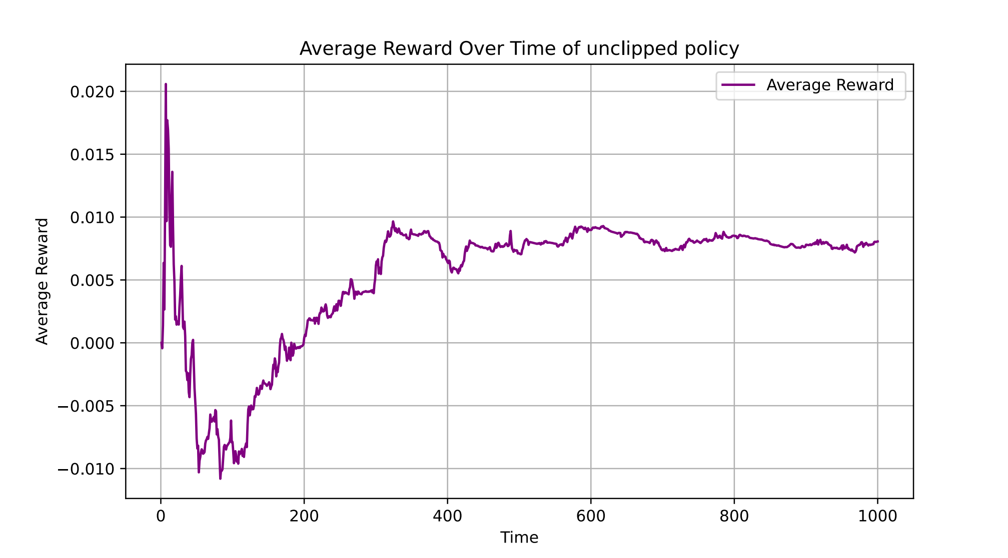

Figure 30: Rewards of π_{cl}

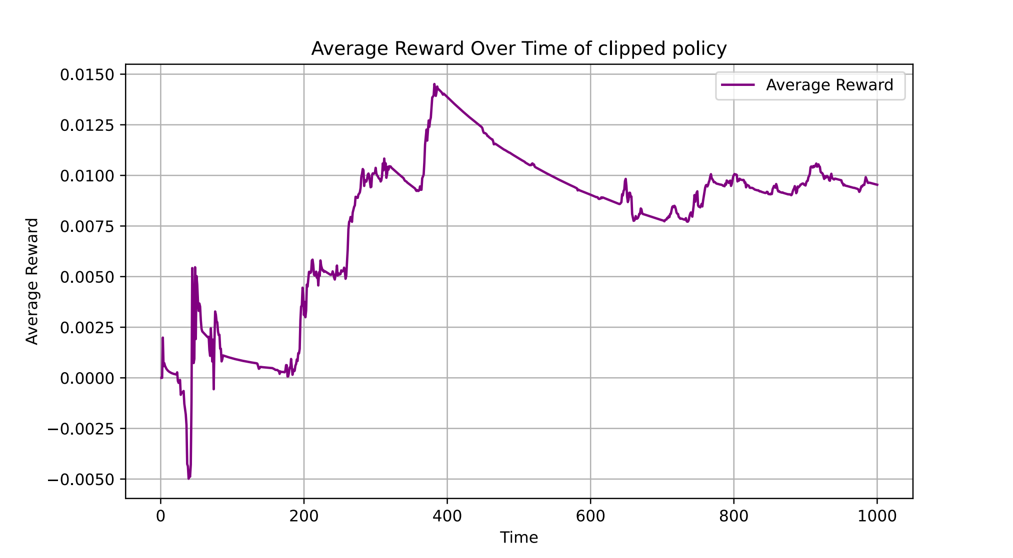

Figure 31: Rewards of π_{pcl}

no good use of this